AOW Technical – Market Model

Contents

What is it?

The Market Model is a prototype for a web based application. It allows us to use the capabilities of the web application to test various market based reports to support ISK making activities in Eve Online.

It has two parts:

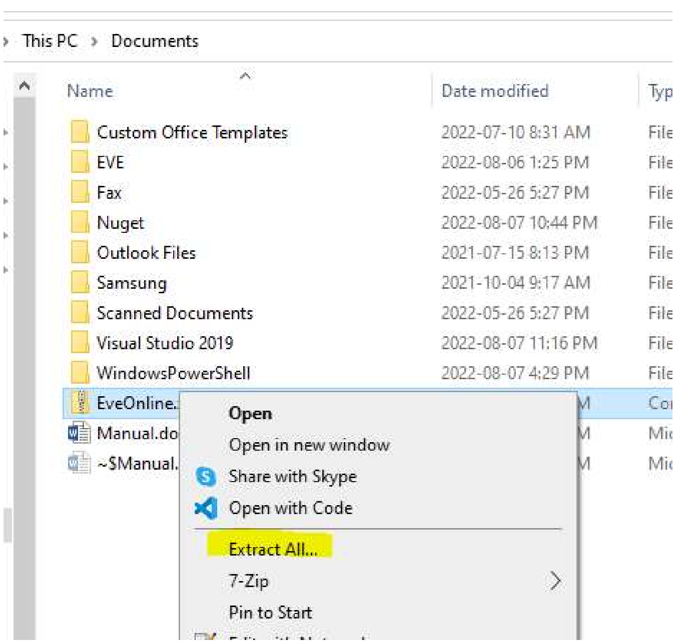
- Data Collection – two tools are provided to collect data from the Eve Swagger API. One tool collects non-personal data (public) and does not require an Eve login. The second tool collects personalized information (private), and hence does require a login to Eve.
- Data Visualization – two spreadsheets are provided. One is for Microsoft Excel, the other can be used by Apache OpenOffice. Both spreadsheets are configured with identical macro-based functions which can be used to extract and use the data collected by the first too.

Installation of the market model

The model can be installed to any folder you wish. Depending on where you "install" it, Powershell may require reconfiguration to allow you to run scripts from that location.

In our example, we will use the \Documents\ folder to make things easier.

Copy the ZIP file to the Documents folder using Windows Explorer. Right-click on the ZIP file and select "Extract All…". Click the "Extract" or "Next" button in the dialog that appears.



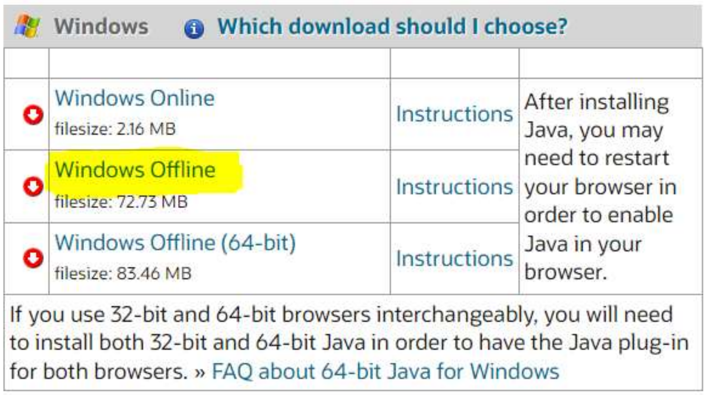
You should have a folder under \Documents named "EveOnline" with all the scripts, data and spreadsheets in it. Delete the ZIP file since it is no longer needed.

Installation of Open Office

Open Office is an open source suite of applications similar to Microsoft Office and is available from the Apache Software Foundation. Unlike Microsoft Office, it is free for personal use.

Before you install it, you need a 32 bit version of Java (even if you are running a 64 bit version of Windows).

The official download page for Java is here: [Java Downloads for All Operating Systems]



Click the highlighted link to download the installer to your \Downloads folder and run the file once it has downloaded. Accept all the defaults until the installer has finished.

The official download page for Open Office is here: [Apache OpenOffice - Official Download]

Click the highlighted link to download the installer to your \Downloads folder and run the file once it has downloaded. Accept all the defaults until the installer has finished.

Security Warning! READ THIS!

The spreadsheets rely heavily on macros. We have not signed them so in order for the spreadsheets to work, you must enable all macros.

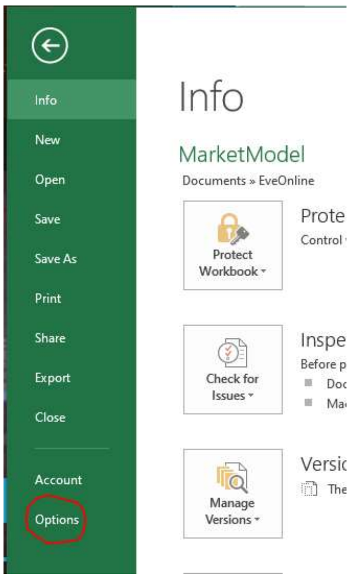
This is risky. For instance, if you receive a file from someone with embedded macros and you enable all macros, it is possible that opening that file you received could infect you with malware. **NEVER open a**

file from someone you do not know, or got from a source you do not trust with settings that enable all macros.

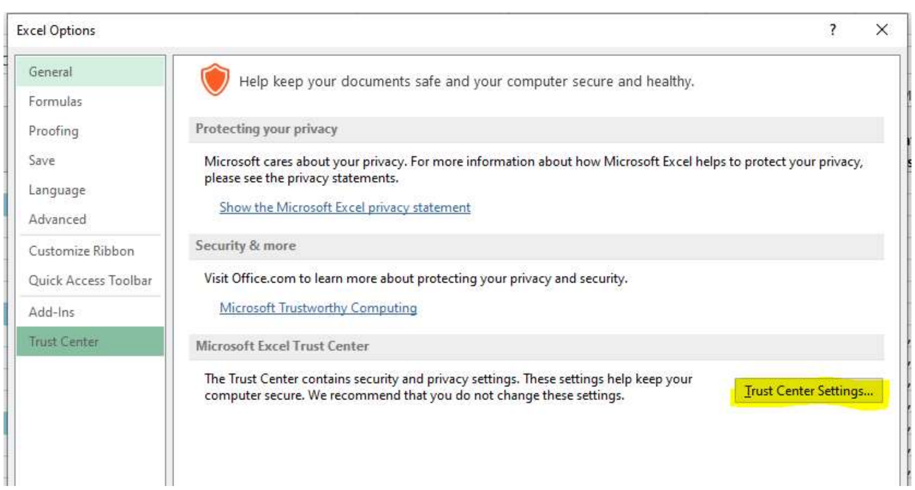
You may choose to enable macros using the following instructions before you open the model, and when finished, reverse the change to ensure you are secure.

Configuration of Excel for macros

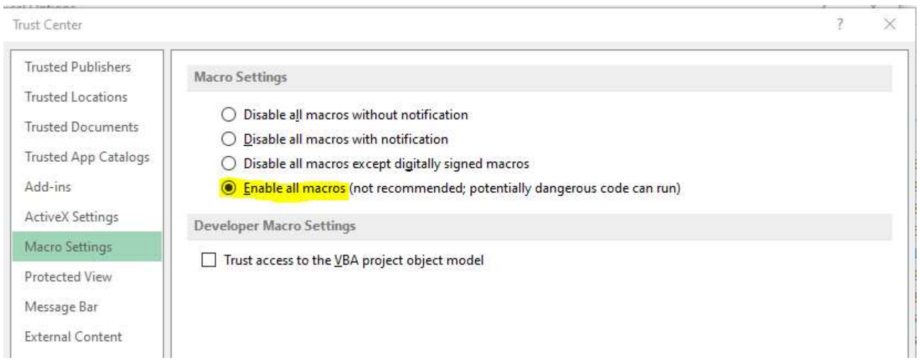
Click on “File” on the menu bar. Click on “Options.”



In the dialog that appears, select “Trust Center” in the left pane and then click the “Trust Center Settings” button.



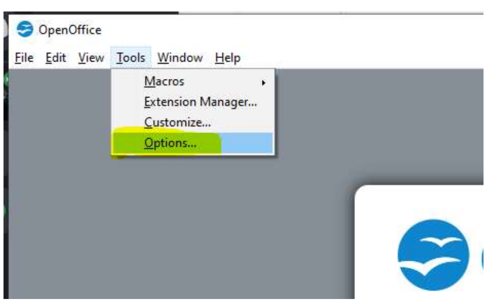
Click on “Macro Settings” in the left pane. Note the settings so you can put them back if security is a concern. Enable all Macros by clicking the highlighted option.



Configuration of Open Office for macros

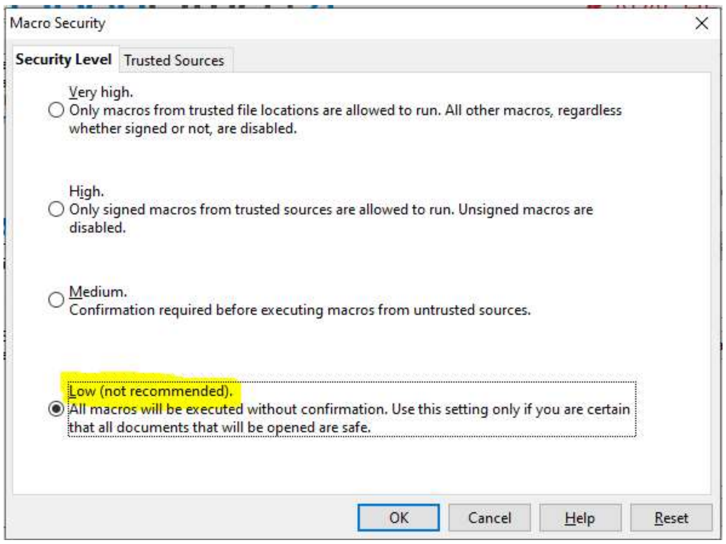
You should have an OpenOffice icon on your desktop, or in your Start Menu. Use it to launch OpenOffice.

From the menu, select “Tools”, “Options.”



In the left pane, click “OpenOffice”, “Security” and click the “Macro Security” button.

Note the settings so you can put them back if security is a concern. Enable all Macros by clicking the highlighted option.



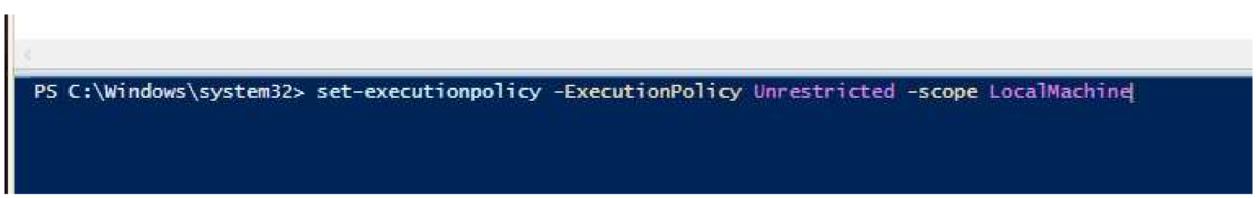
Data Collection

The data collection scripts are written in Windows Powershell, a very powerful scripting language that is installed on every Windows computer since Windows 7. This means that no installation is necessary in order to use the scripts.

There are several security settings that may need to be enabled to allow the script(s) to run. You should only have to change them the first time you run the scripts. In order to change the security settings in Powershell, you must run it as administrator.

You can find Powershell in the Start Menu under “Windows Powershell”, locate the “Windows Powershell ISE” entry, right-click on it and choose “More”, then “Run as Administrator.”

In the blue window (the console), type “set-executionpolicy -ExecutionPolicy Unrestricted -scope LocalMachine” and hit Enter.



In the dialog that pops up, click “Yes to All” and close the Powershell ISE.

** Edits pending here, file may need to be unblocked.”

Input Files

The script saves its data on Comma Separated Value (.csv) files. This makes them very easy to import to a spreadsheet or a database.

There are two mandatory files that control the behaviour of the script:

WatchMarkets.csv – this file specifies the regions and markets to save the pricing and order data from. It is possible to add or remove entries and this either increases or reduces the overall script run time and the amount of data generated.

Columns are (in order):

1. Market ID – this is a numerical identifier for the station or structure that hosts the market. The ID can usually be found in one of two other CSV files: Structures.csv (player stations) or Stations.csv (NPC stations.) If this data is missing, the script will attempt to locate the market by the name and add the Market ID to the file upon completion.
2. Name – this is a short version for the market name and is used by the spreadsheet for data lookups. For instance "Jita" for "Jita IV - Moon 4 - Caldari Navy Assembly Plant."
3. Station Name – this is the long version of the market name, "Jita IV - Moon 4 - Caldari Navy Assembly Plant."
4. Region Name – this is the name of the region and is used for display in the progress bar when data is being downloaded, "The Forge."
5. Region ID – this is the numerical identifier for the region, it can be found in the Regions.csv file.

WatchItems.csv – this file specifies the item prices and orders you wish to track. It is possible to add or remove entries and this either increases or reduces the overall script run time and the amount of data generated.

Columns are (in order):

1. Item ID – this is a numerical identifier for the product. It can be found in the Types.csv file. For instance, Plagioclase has an item id of 18.
2. Item Name – this is the full name of the product, for example: "Plagioclase."

The following files are optional. If they do not exist, the script will query Eve for a full update, saving the data for future use and updates. If they exist, they will be read first and any changes that are found (new stations, items, etc.) will be made and saved over time.

Regions.csv – this is a list of regions with their ID's.

Columns are (in order):

1. Region ID – the numerical identifier for the region, for example: 10000001
2. Region Name – the full name of the region, for example: Derelik

Stations.csv – this is a list of every NPC station in Eve with its associated numerical identifier and the identifier for the system it is located in.

Columns are (in order):

1. Station ID – the numerical identifier for the station, for example: 60003760
2. Station Name – the full name of the station, for example: Jita IV - Moon 4 - Caldari Navy Assembly Plant
3. System ID – the numerical identifier for the system the station is in, for example: 30000142

Structures.csv – this is a list of every “open” player structure, or if updated, every structure you have access to with its associated numerical identifier and the identifier for the system it is located in.

Columns are (in order):

1. Structure ID – the numerical identifier for the structure, for example: 1036974821465
2. Structure Name – the full name of the structure, for example: Jel - Jel clone center
3. System ID – the numerical identifier for the system the structure is in, for example: 30005222

Systems.csv – this is a list of every system in Eve with its numerical ID, name and security level.

Columns are (in order):

1. System ID – the numerical identifier for the system, for example: 30005222
2. System Name – the full name of the system, for example: Serren
3. Security – the numerical identifier for the system security, for example: 0.4588785171508789

Types.csv – every item in Eve has a name and an ID, this file contains all known items and their ID’s.

Columns are (in order):

1. Type ID – the numerical identifier for the item, for example: 18
2. Type Name – the full name of the item/type, for example: Plagioclase

Output Files

The script produces two files with new content, the content from previous runs are lost.

Orders.csv – this is a list of every observed order at the markets listed in WatchMarkets.csv, and only for the items listed in WatchItems.csv.

Columns are (in order):

1. Type ID – the numerical identifier for the item, for example: 18
2. Station ID – the station ID for where the order is at, for example: 60003760
3. Transaction Type – Buy or Sell
4. Price – the buy or sell price in ISK, each
5. Total – the total order volume in items
6. Remaining – the remaining number of items for buy or sale

Prices.csv – this is a list of statistical observations based on the prices data.

Columns are (in order):

1. Type ID – the numerical identifier for the item, for example: 18
2. Station ID – the station ID for where the order is at, for example: 60003760
3. Transaction Type – Buy or Sell
4. High – the highest price observed for the item/transaction type in ISK
5. Low – the lowest price observed for the item/transaction type in ISK
6. Average – the average price per unit observed for the item/transaction type in ISK

Collecting the Data

There are two types of data available in the Eve API – public and private. All information that is available can be browsed here: [EVE Swagger Interface (evetech.net)]

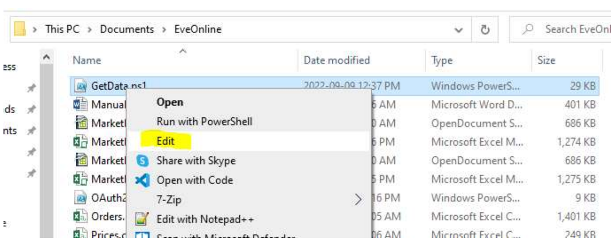
As you browse the data, you will see that some data has a “lock” icon and some data does not. This “lock” icon indicates that the data is not public and you have to log in to Eve to retrieve it.

The non-public data is personalized and different for each player.

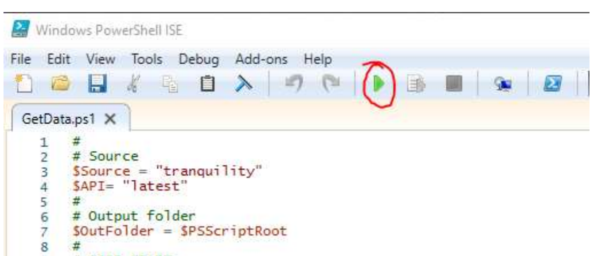
Public Data Refresh (GetData.ps1)

In most cases, you only need to update the public data. The scripts are separated by the type of data they collect to allow collection of data without the need to log in which can be improved with a scheduled and automated data refresh. At this time, all refreshes are manual.

To run the script, open the EveOnline folder in Windows Explorer, select the GetData.ps1 file. Right click on it and choose “Edit.” This will open the script in the Powershell ISE.

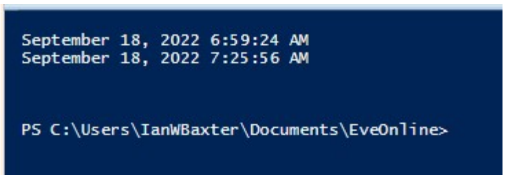


The little green “arrow” on the ribbon bar is the “Run” button. Click it.



As the script executes, a progress bar will be displayed as the order data is downloaded for each configured regional market.

Once the script completes, the blue area (console) will look like this, you can close the Powershell ISE like you would any other Windows application. The two date/time entries indicate when the script started and finished.



The updated Price and order data is now ready to be updated in whichever data visualization tool you have chosen (Excel or OpenOffice Calc.)

Private Data Refresh (Oauth2.ps1)

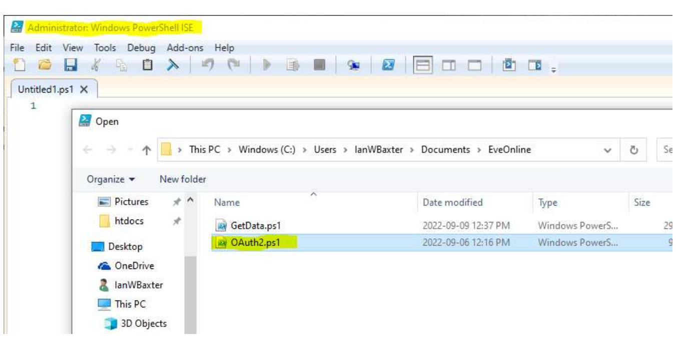
In order to access character-specific data in Eve such as the Structures data, you must provide access to the requested data by signing in to Eve. This is called "Single Sign On" and the process is documented here: [Single Sign On (SSO) – EVE Online]

Essentially, the script creates a secure, random key, opens your web browser, sends you to the Eve sign on page and passes the sign on page the secure key. Once you have signed in, the Eve sign on page will redirect you to a web server running on your own computer with a verification code that can be used to ensure that nobody has "spoofed" your credentials. The process of creating and running a web server on your computer requires that you have administrative rights so a different process is required to run this script.

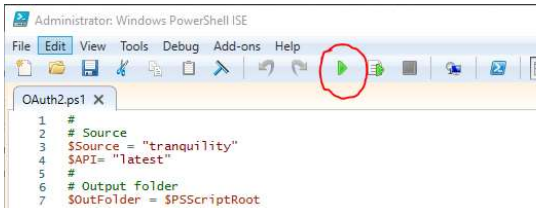
At no time is your Eve Online account information captured or stored by the script, it is impossible to do so. This is also the same method most websites use to allow sign-in via your Facebook, Google, Microsoft or other accounts. Read the link above to understand how to verify that you are using a valid login page for Eve online. This same technique can be used for other single sign on providers.

You can find Powershell in the Start Menu under "Windows Powershell", locate the "Windows Powershell ISE" entry, right-click on it and choose "More", then "Run as Administrator."

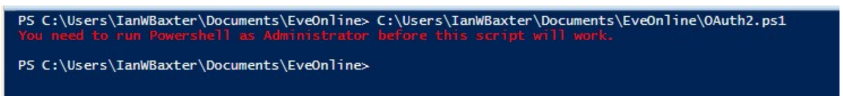
If you did it correctly and approved the application to run as an administrator, Powershell ISE will open. Note that the title will show as "Administrator: Windows Powershell ISE". Click "File", "Open" and select the OAuth2.ps1" script.



The little green “arrow” on the ribbon bar is the “Run” button. Click it.

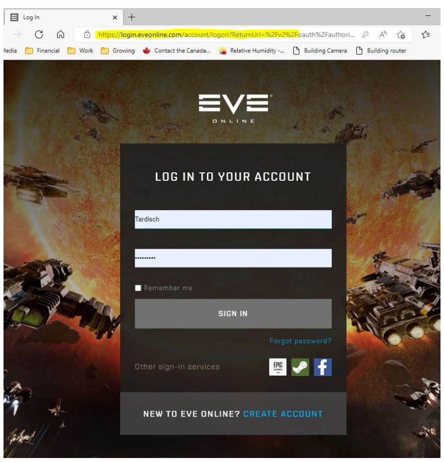


If you did not launch Powershell ISE as an Administrator, the following message will be displayed in the console:

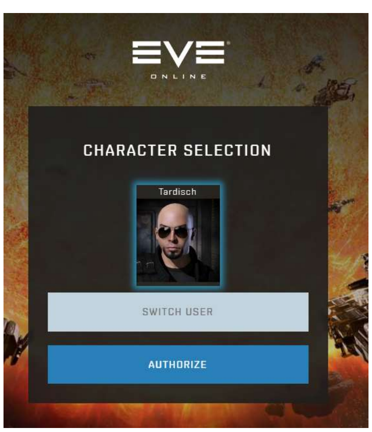


If you have a web server running on your computer, you will see a lot of errors in the Powershell console. You need to stop the web server service for it to be able to create its own web server.

If everything worked correctly, you should see the Eve Online sign in page with no errors in the Powershell console.

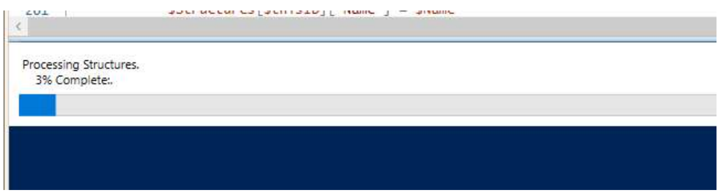


Log in to Eve and then select the character whose structure data you wish to update. Click the “Authorize” button.



Your web browser will now be redirected to the temporary web server running on your computer with the authorization code. You can close this window.

The progress bar will display the script's progress. When the progress bar clears, the script has finished.



Data Visualization

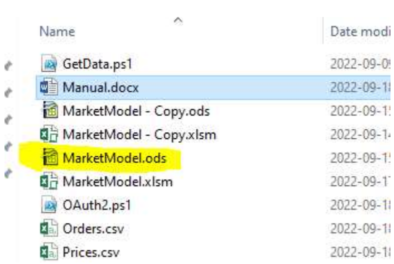
Now that we have the market data, it is time to do something with it. Several possibilities for uses are created as example worksheets in the Excel and OpenOffice models. They are almost identical except for one formula and one automated process which proved too difficult to translate from Excel.

Updating the Prices

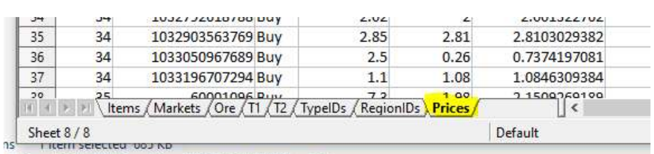
Loading the pricing data into the model is the automation process which could not be made identical between the two model versions.

Open Office

First, open the model – using Windows Explorer, open the Documents\EveOnline folder, locate the OpenOffice file 'Marketmodel.ods' by name, or if you do not see the file extension, then by name and the icon. Use the snapshot below for reference. Double-click on it to open it.

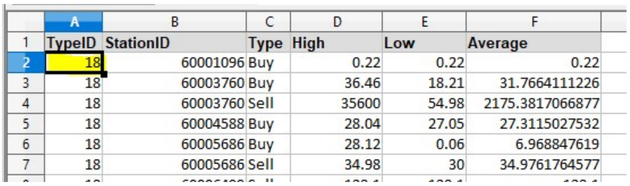


At the bottom of the model window are a bunch of tabs, these are worksheets. Click on the tab for the Prices worksheet.



Make sure you are at the top of the sheet data by using the scrollbar at the right of the window. Note that above the sheet data are letters indicating the column name, and numbers at the left to indicate the row number. The two in combination, “A2” give the “address” of the box, or cell.

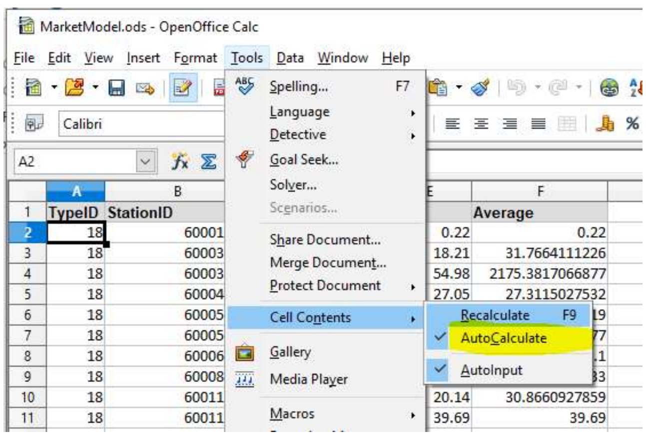
Click in cell “A2”, in the snapshot below:



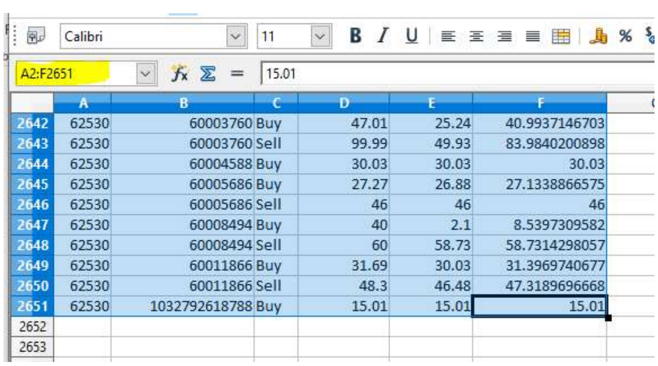
	A	B	C	D	E	F
1	TypeID	StationID	Type	High	Low	Average
2	18	60001096	Buy	0.22	0.22	0.22
3	18	60003760	Buy	36.46	18.21	31.7664111226
4	18	60003760	Sell	35600	54.98	2175.3817066877
5	18	60004588	Buy	28.04	27.05	27.3115027532
6	18	60005686	Buy	28.12	0.06	6.968847619
7	18	60005686	Sell	34.98	30	34.9761764577

Before we remove that price data and replace it, we need to make sure that the model will not try to update while we are doing the work. If we do not, then it will take longer to update the data.

From the menu, select “Tools”, “Cell Contents” and click on “AutoCalculate” to remove the check mark to the left, indicating automatic calculation has been turned off.

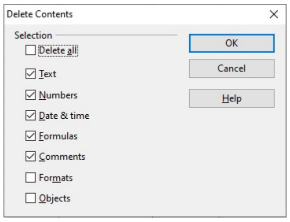


Select the data only by selecting cell “A2” and then hitting the key combination Shift-Ctrl-End. The display will change to show you have selected to the end of the data. Note the highlighted addresses in the snapshot. Two addresses with a colon between them specifies an area, or “range.”

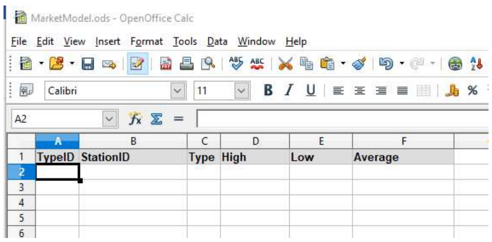


	A	B	C	D	E	F
2642	62530	60003760	Buy	47.01	25.24	40.9937146703
2643	62530	60003760	Sell	99.99	49.93	83.9840200898
2644	62530	60004588	Buy	30.03	30.03	30.03
2645	62530	60005686	Buy	27.27	26.88	27.1338866575
2646	62530	60005686	Sell	46	46	46
2647	62530	60008494	Buy	40	2.1	8.5397309582
2648	62530	60008494	Sell	60	58.73	58.7314298057
2649	62530	60011866	Buy	31.69	30.03	31.3969740677
2650	62530	60011866	Sell	48.3	46.48	47.3189696668
2651	62530	1032792618788	Buy	15.01	15.01	15.01
2652						
2653						

Hit the “Delete” key and make sure you have selected the options shown below, click “OK.”

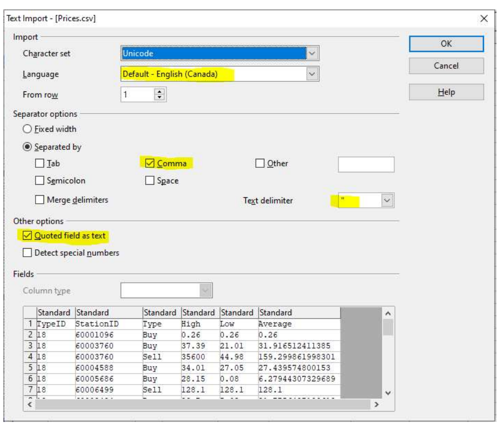


Now hit the key combination Ctrl-Home to go back to the top of the sheet. Select cell “A2” again.

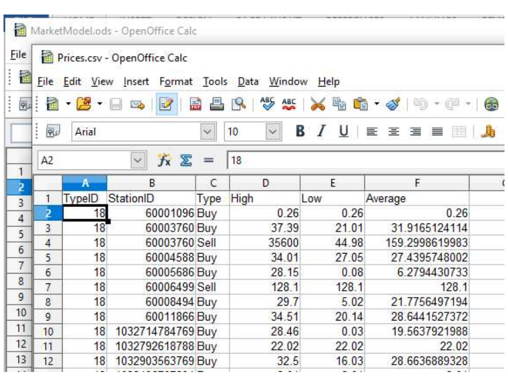


Using Windows Explorer, locate the Prices.csv file (your icon will appear different) and double-click on it to open it if OpenOffice Calc.

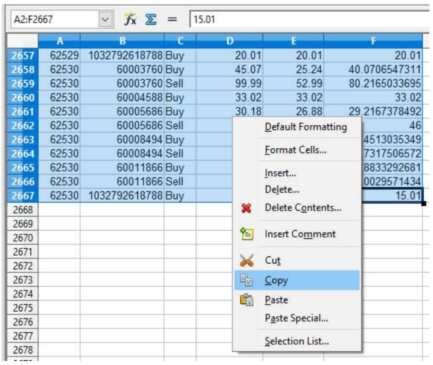
A dialog box will appear. Make sure that the Language is set correctly for your location, the comma is set as a separator, the text delimiter is double quotes and the quoted field as text option is selected. Click “OK.”



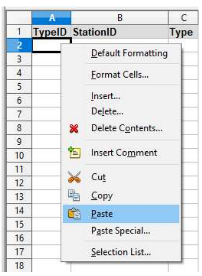
Now that you have the Prices.csv file open, select Cell “A2” from that sheet.



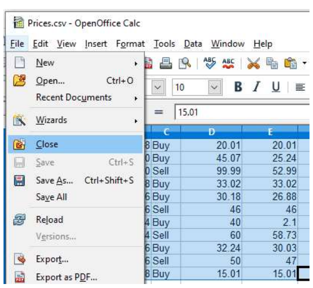
Press Shift-Ctrl-End to select all the data. This time, right-click on the blue area and select “Copy” from the menu that appears.



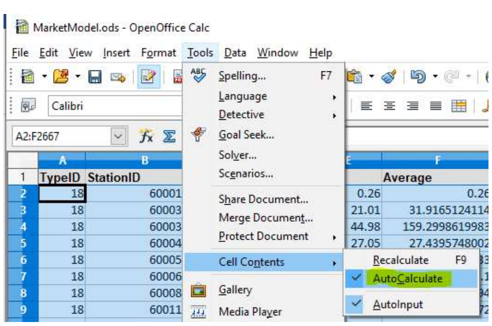
Switch to the “MarketModel.ods - OpenOffice Calc” window, select cell “A2” and right click on it. Select “Paste” from the menu that appears.



Switch to the “Prices.csv – OpenOffice Calc” window and close it.

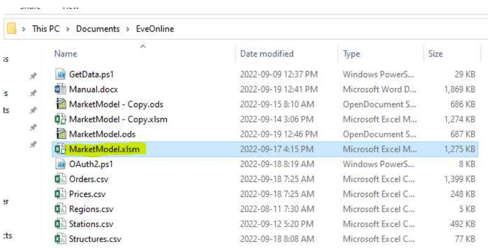


Switch to the “MarketModel.ods - OpenOffice Calc” window. From the menu, select “Tools”, “Cell Contents” and click on “AutoCalculate” to force the formulas in the model to update.



Excel

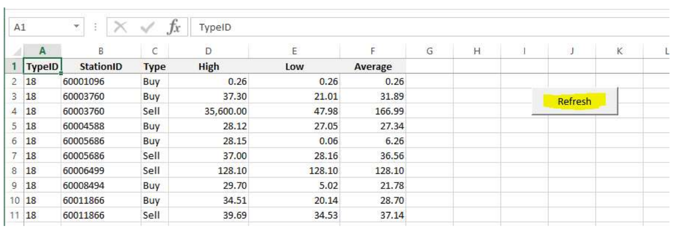
First, open the model – using Windows Explorer, open the Documents\EveOnline folder, locate the OpenOffice file 'Marketmodel.xlsm' by name, or if you do not see the file extension, then by name and the icon. Use the snapshot below for reference. Double-click on it to open it.



At the bottom of the model window are a bunch of tabs, these are worksheets. Click on the tab for the Prices worksheet.

Make sure you are at the top of the sheet data by using the scrollbar at the right of the window. Note that above the sheet data are letters indicating the column name, and numbers at the left to indicate the row number. The two in combination, "A2" give the "address" of the box, or cell.

Click the "Refresh" button and wait for the sheet to finish calculating.



Customizing the Model

The intent of this document is to teach you how to model data from the Eve Online markets so some basic spreadsheet knowledge is required from this point onwards. Instructions will not be as detailed and there will be fewer step by step instructions.

Our focus will be on "how can I make this help me read the markets and not so much on how to do it procedurally.

Items Sheet – Watching more prices in the markets

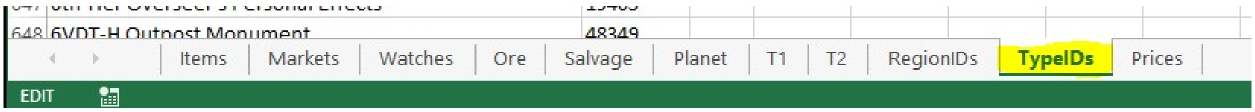
The Items sheet can be used to track and verify prices for items in the WatchItems.csv file, a main configuration component of the Market Model.

This sheet can also be used to support hauling operations, basically buying items in one market, then transporting them and selling them in another for profit.

It can also be used to check market prices for items looted from NPC's from ratting operations.

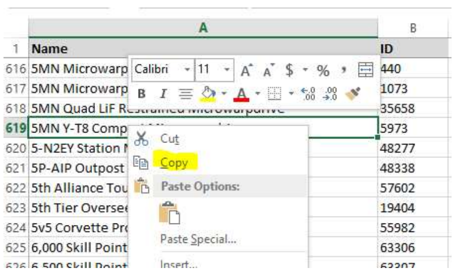
Since the data in the sheet is not a complete list of items in Eve, from time to time you may wish to add additional items. In this example, we will add a common item that drops from NPC's – the "5MN Y-T8 Compact Microwarpdrive."

You can find the information you need in the TypeIDs sheet.



The data in the TypeIDs sheet is sorted so you can just scroll down to locate cells starting with "5MN", or use the find feature in the spreadsheet application.

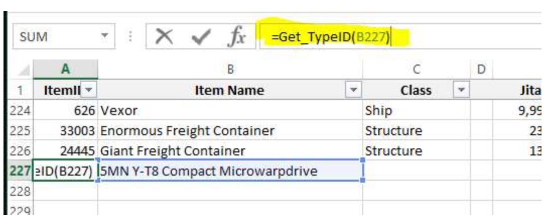
You only need to select and copy the item name.



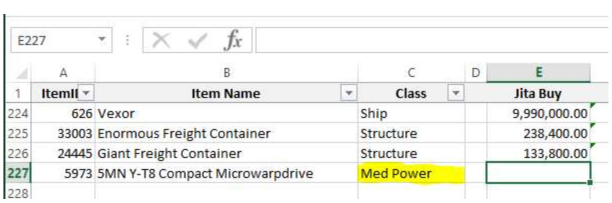
Switch to the Items sheet, scroll down to the end of the list and paste the name into the first blank cell.

The ItemID is necessary for proper operation of our custom formulas. In this case, we pasted the name in cell B227, so select cell A227 and type the following formula into it: =Get_TypeID(B227)

This will look up the TypeID for the item name in cell B227 and return it in cell A227.

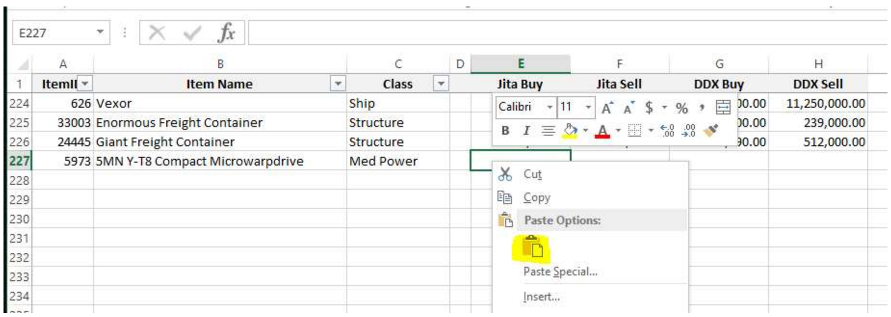


I use the "Class" data to group items by function, in this case it fits into a medium power slot, so the class should be "Med Power". You can use any value you want in this cell (C227).

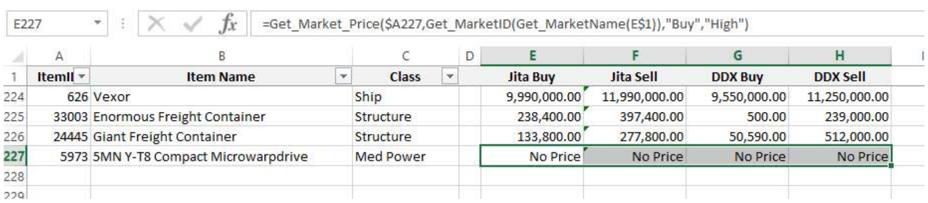


Finally, we need to add the formulas to look up the item prices from the Prices sheet. Select the price data from the row above (range E226:H226), right click on the selection and choose "Copy."

Now, right-click in cell E227 and choose "Paste."

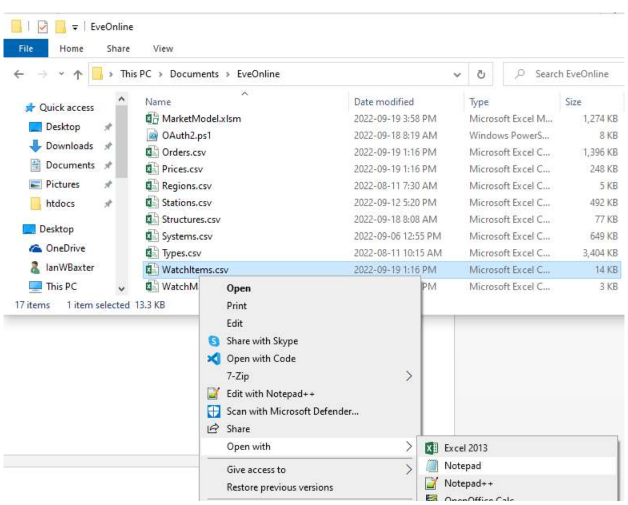


The data will update and you will now see "No Price" displayed in in the market price data.

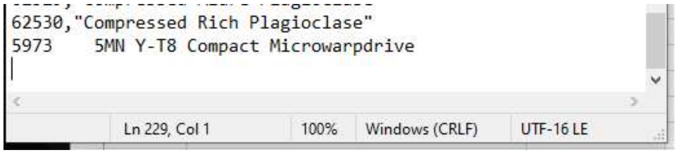


This is because the WatchItems.csv file needs to be configured to collect the price data for that item. This is an easy thing to do. First, open the WatchItems.csv file in a text editor like Notepad.

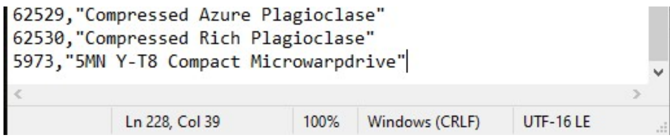
To do this, use Windows Explorer, find and right-click on the file. Choose "Open with…" and then Notepad.



In Notepad, hit the key combination Ctrl-End to go to the end of the file. Switch to the Market Model and copy the data from the Items sheet columns A and B (the ID and name of the item). Paste it at the end of the file data in Notepad.

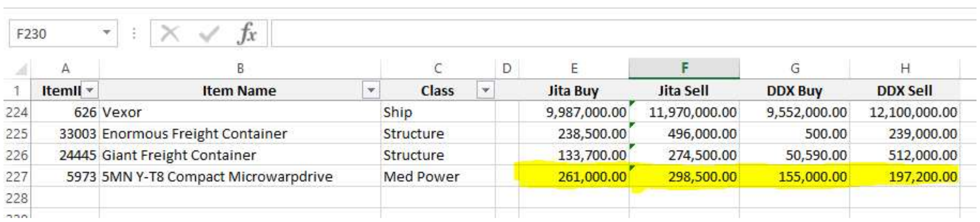


The blank space between the ID and the name is a tab character, change that to a comma and make sure you also put double quotes around the item name.



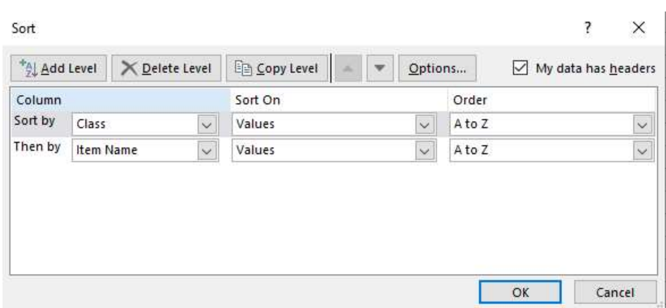
Save the notepad file and close Notepad.

You now need to update the prices data using the Powershell script and then import the prices.csv data into the model. Once this is done, you should see prices listed in the Items sheet.



	ItemI	Item Name	Class	D	Jita Buy	Jita Sell	DDX Buy	DDX Sell
224	626	Vexor	Ship		9,987,000.00	11,970,000.00	9,552,000.00	12,100,000.00
225	33003	Enormous Freight Container	Structure		238,500.00	496,000.00	500.00	239,000.00
226	24445	Giant Freight Container	Structure		133,700.00	274,500.00	50,590.00	512,000.00
227	5973	5MN Y-T8 Compact Microwarpdrive	Med Power		261,000.00	298,500.00	155,000.00	197,200.00
228								

Hint: You can “organize” this sheet by using the spreadsheet application to sort the data. It is recommended to sort by “Class” first and then by “Item Name.”



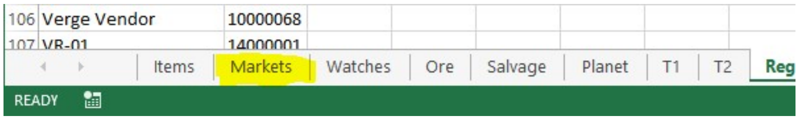
Markets Sheet – Watching more markets

The Market Model will only watch prices in markets you specify. You can add or remove markets and have multiple markets in the same region. The regional market data is only queried once during the order download process in Powershell.

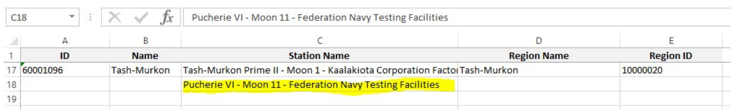
In this example, we will add a new market and configure it in Watchmarkets.csv

We will use a market in Sinq Liason, an NPC station (so it is in Stations.csv) named “Pucherie VI - Moon 11 - Federation Navy Testing Facilities.”

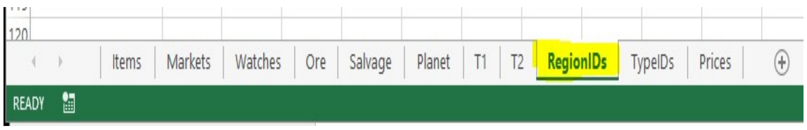
Switch to the “Markets” sheet.



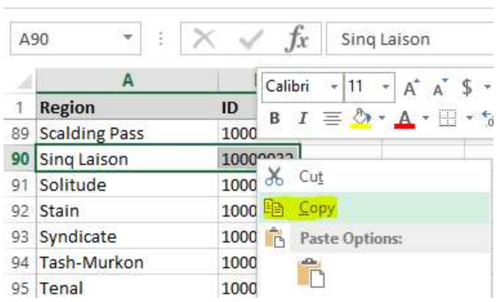
Enter the station name at the end of the data in column “C.”



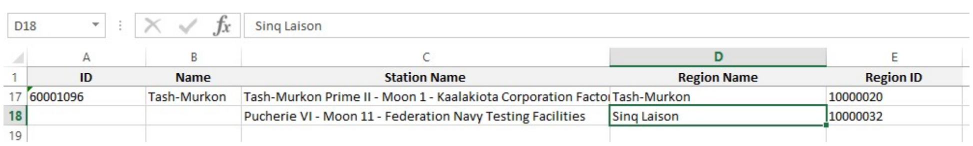
Switch to the RegionIDs sheet.



Find the entry for Sinq Liason and copy the region name and ID.



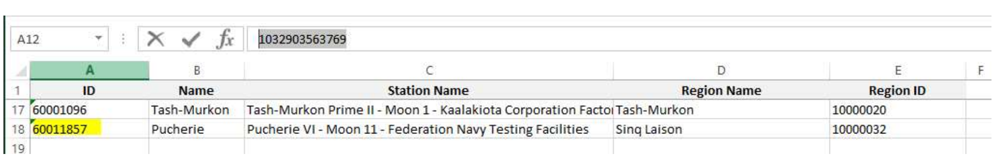
Paste in in the Markets sheet in the cell for the Region name.



Enter Pucherie in the “Name” column. This is an “alias” we will use for that station.

You can find the Station ID (we call it Market ID) in the stations.csv file. It is the first item in the line. Note that NPC stations use ID’s starting with 6000000 where player structures (in structures.csv) start with 1000000000000.

Paste the Station ID in column “A” of the Markets sheet.



Now we have to add the market to the WatchMarkets.csv file. Open it in Notepad (or your favorite text editor) and hit Ctrl-End to go to the last line. Switch back to the Markets sheet and copy the data for the market and paste it to the end of WatchMarkets.csv

```
*WatchMarkets.csv - Notepad
File  Edit  Format  View  Help
60001096,"Tash-Murkon","Tash-Murkon Prime II - Moon 1 - Kaalakiota Corporation Factory","Tash-Murkon",10000020
60006499,"Teonusude","Teonusude III - Moon 8 - Imperial Armaments Factory","Molden Heath",10000028
1032792618788,"Villore","Villore - IChooseYou Market and Industry","Essence",10000064
60011857	Pucherie	Pucherie VI - Moon 11 - Federation Navy Testing Facilities	Sinq Laison	10000032
```

Replace the tabs with commas and make sure that the text data is enclosed in double quotes. Save the file when you are done.

```
WatchMarkets.csv - Notepad
File  Edit  Format  View  Help
1032714784769,"Sobaseki","Sobaseki - IChooseYou Market and Industry","LoneTrek",10000016
60001096,"Tash-Murkon","Tash-Murkon Prime II - Moon 1 - Kaalakiota Corporation Factory","Tash-Murkon",10000020
60006499,"Teonusude","Teonusude III - Moon 8 - Imperial Armaments Factory","Molden Heath",10000028
1032792618788,"Villore","Villore - IChooseYou Market and Industry","Essence",10000064
60011857,"Pucherie","Pucherie VI - Moon 11 - Federation Navy Testing Facilities","Sinq Laison",10000032
```

Run the public data collection script again and update the prices data in the model. You will need it for the next section.

Items sheet – “Hauling” model (Adding markets, math, conditional formatting)

In addition to providing a backup for the WatchItems.csv file, the Items data can be used to identify “market opportunities” for buying items in one market and then re-selling them in another at a profit.

The limitation of the model is that it is not real-time data and by the time you get to the source or destination, the opportunity could have been lost. Market browsers such as Eve Tycoon also have this limitation since the data is not real-time. Nothing is guaranteed in Eve, but the model is better than guessing.

The Items sheet is already set up for Dodixie and Jita, listing Buy and Sell order prices for each. If you plan to sell immediately, you would be selling items to fill a Buy order with the highest price in that case. If you choose to place items for sale over time, then you would be looking at the lowest Sell order price.

B19 | 125mm Railgun II

	A	B	C	D	E	F	G	H	I
1	ItemI	Item Name	Class		Jita Buy	Jita Sell		DDX Buy	DDX Sell
2	230	Antimatter Charge M	Charge		59.07	66.90		35.01	58.97
3	225	Iridium Charge M	Charge		32.02	59.77		6.00	45.50
4	237	Plutonium Charge L	Charge		49.54	115.80		No Price	138.00
5	229	Plutonium Charge M	Charge		45.26	72.94		45.30	79.90
6	221	Plutonium Charge S	Charge		11.30	18.98		6.86	19.20
7	224	Tungsten Charge M	Charge		24.07	29.45		1.44	38.49
8	228	Uranium Charge M	Charge		51.00	64.00		13.13	54.99

When you sell items on the market, you also need to pay a broker’s fee and taxes. Broker fees can be reduced by training the Broker Relations skill, and taxes can be reduced by training the Accounting skill. As a general rule, broker fees and taxes are usually around 10%, but can be reduced to 6.6% by fully training those skills. This means that you must make about 10% profit to break even, and more than 10% to actually profit.

Profit percentage can be calculated as (((<Sell Price> – <Taxes and Fees>)/<Buy Price>)-1) *100

In the example above, it looks like there is an opportunity buying “Antimatter Charge M” in Dodixie for 58.97 ISK each and then selling in Jita over time at 66.75 ISK (just below the lowest Sell order at 66.90 ISK). Taking into account for taxes and fees, you would get 60.075 ISK (66.75*.9) selling at 66.75 ISK.

The formula then would be (((66.75*.9)/58.97)-1)*100, or 1.87% profit. With the effort involved and the risk of being undercut selling over time in Jita, it seems hardly worth the time, yet people do it and lose money.

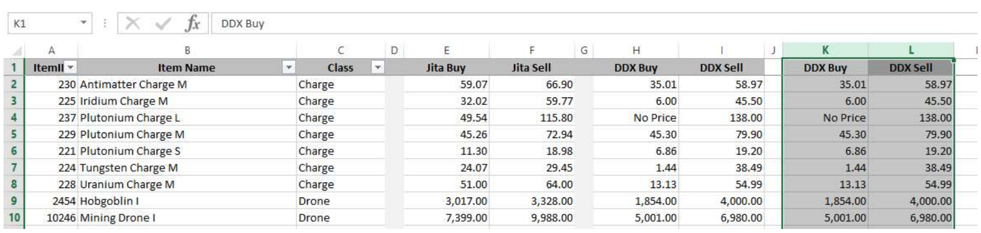
Using this example, we will add another market to the Items sheet. There is always a greater profit potential selling items in non-hub markets. The market we just added in the last section, Pucherie, is just such a market.

Select the entire columns “H” and “I”, right-click on them and select “Copy” from the dropdown menu.


H1 | DDX Buy

	ItemI	Item Name	Class	Jita Buy	Jita Sell	DDX Buy	DDX Sell
2	230	Antimatter Charge M	Charge	59.07	66.90	35.01	58.97
3	225	Iridium Charge M	Charge	32.02	59.77	6.00	45.5
4	237	Plutonium Charge L	Charge	49.54	115.80	No Price	138.(
5	229	Plutonium Charge M	Charge	45.26	72.94	45.30	79.9
6	221	Plutonium Charge S	Charge	11.30	18.98	6.86	19.2
7	224	Tungsten Charge M	Charge	24.07	29.45	1.44	38.4
8	228	Uranium Charge M	Charge	51.00	64.00	13.13	54.9
9	2454	Hobgoblin I	Drone	3,017.00	3,328.00	1,854.00	4,000.(

Select Cell “K1”, right-click in it and paste the data you just copied. The sheet will recalculate.


K1 | DDX Buy

	ItemI	Item Name	Class	Jita Buy	Jita Sell	DDX Buy	DDX Sell	DDX Buy	DDX Sell
2	230	Antimatter Charge M	Charge	59.07	66.90	35.01	58.97	35.01	58.97
3	225	Iridium Charge M	Charge	32.02	59.77	6.00	45.50	6.00	45.50
4	237	Plutonium Charge L	Charge	49.54	115.80	No Price	138.00	No Price	138.00
5	229	Plutonium Charge M	Charge	45.26	72.94	45.30	79.90	45.30	79.90
6	221	Plutonium Charge S	Charge	11.30	18.98	6.86	19.20	6.86	19.20
7	224	Tungsten Charge M	Charge	24.07	29.45	1.44	38.49	1.44	38.49
8	228	Uranium Charge M	Charge	51.00	64.00	13.13	54.99	13.13	54.99
9	2454	Hobgoblin I	Drone	3,017.00	3,328.00	1,854.00	4,000.00	1,854.00	4,000.00
10	10246	Mining Drone I	Drone	7,399.00	9,988.00	5,001.00	6,980.00	5,001.00	6,980.00

In cells “K1” and “L1” change “DDX” to “Pucherie”. After the recalculation, you now have Pucherie prices listed.


L2 | =Get_Market_Price($A2,Get_MarketID(Get_MarketName(L$1)),"Sell","Low")

	ItemI	Item Name	Class	Jita Buy	Jita Sell	DDX Buy	DDX Sell	Pucherie Buy	Pucherie Sell
2	230	Antimatter Charge M	Charge	59.07	66.90	35.01	58.97	No Price	No Price
3	225	Iridium Charge M	Charge	32.02	59.77	6.00	45.50	No Price	No Price
4	237	Plutonium Charge L	Charge	49.54	115.80	No Price	138.00	No Price	No Price
5	229	Plutonium Charge M	Charge	45.26	72.94	45.30	79.90	No Price	No Price
6	221	Plutonium Charge S	Charge	11.30	18.98	6.86	19.20	No Price	No Price
7	224	Tungsten Charge M	Charge	24.07	29.45	1.44	38.49	No Price	No Price

Notice the formula in the formula bar, this is really what is doing all the work. If you click inside this formula, the spreadsheet will highlight the cells it uses for easier identification.


SUM | =Get_Market_Price($A2,Get_MarketID(Get_MarketName(L$1)),"Sell","Low")

	ItemI	Item Name	Class	Jita Buy	Jita Sell	DDX Buy	DDX Sell	Pucherie Buy	Pucherie Sell
2	230	Antimatter Charge M	Charge	59.07	66.90	35.01	58.97	No Price	=Get_Market_Pr
3	225	Iridium Charge M	Charge	32.02	59.77	6.00	45.50	No Price	No Price
4	237	Plutonium Charge L	Charge	49.54	115.80	No Price	138.00	No Price	No Price

How this works:

=Get_Market_Price($A2,Get_MarketID(Get_MarketName(L$1)),"Buy","High")

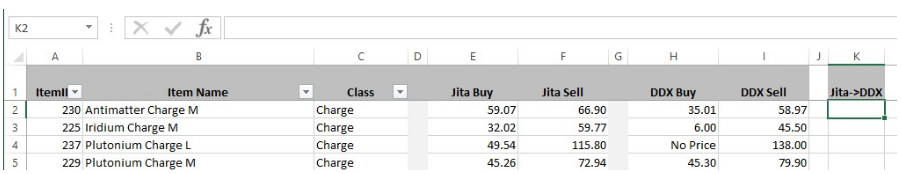
1. The market name, “Pucherie” is collected from cell L$1 with the Get_Market_Name custom function.
2. The market ID is collected from the market name “Pucherie” with the Get_Market_ID custom function.
3. The value in cell “$A2” is substituted for the cell address
4. The custom function call Get_Market_Price then looks like this: =Get_Market_Price(230, 60011857,”Buy”,”High”)
5. The Get_Market_Price custom function looks up the data for the item, market id and order type on the Prices sheet to return the high price back to the sheet.

Order types that can be specified are: “Buy” or “Sell.” Price data can be “High”, “Low” or “Average.”

Pucherie doesn’t seem to be a very active market. Remove it from the Items sheet by selecting columns “K” and “L”, right-clicking on the selection and choosing “Delete” from the dropdown menu.

Also delete the row in the Markets sheet for Pucherie, and the line holding Pucherie data from Watchmarkets.csv to save time downloading the prices data.

A profit calculation can be added to the Items sheet to make it easier to identify profit opportunities. In this case, we will see if we can buy stuff in Jita and sell it over time for profit in Dodixie. Select cell “K1” and enter Jita->DDX so you know what data we will display in that column.

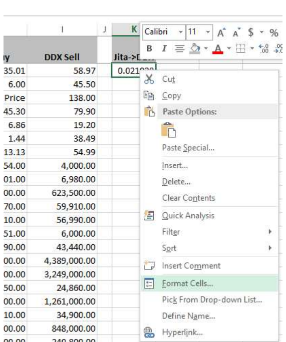


ItemI	Item Name	Class		Jita Buy	Jita Sell		DDX Buy	DDX Sell		Jita->DDX
230	Antimatter Charge M	Charge		59.07	66.90		35.01	58.97		
225	Iridium Charge M	Charge		32.02	59.77		6.00	45.50		
237	Plutonium Charge L	Charge		49.54	115.80		No Price	138.00		
229	Plutonium Charge M	Charge		45.26	72.94		45.30	79.90		

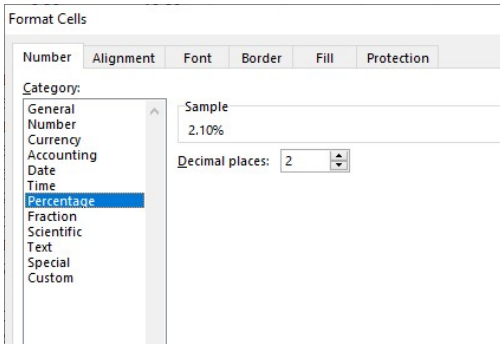
In cell “K2” enter the following formula: =(($I2*0.9)/$F2)-1” (without the quotes)

Note that we have a dollar sign before the column portion of the cell address. This tells the spreadsheet that if the formula is pasted into another cell that this “column” part of the cell address should not change. You can do this with the row component of the address (ie. A$2), or the entire cell address (ie. A2.)

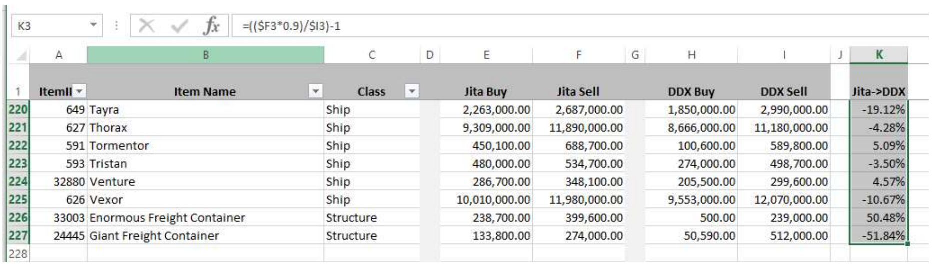
Right-Click on cell “K2” and select “Format cells” from the dropdown menu.



Select the option to format the cell as “Percentage” with two decimal points.



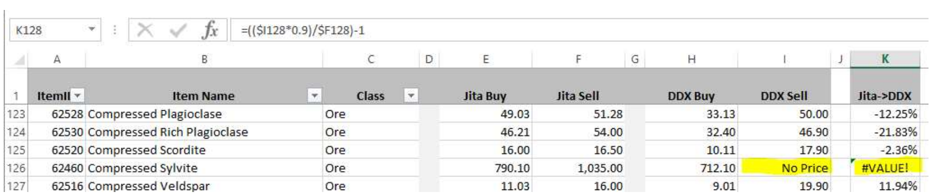
Copy and paste the formula down for all the rows with data.



	ItemI	Item Name	Class	Jita Buy	Jita Sell	DDX Buy	DDX Sell	Jita->DDX
220	649	Tayra	Ship	2,263,000.00	2,687,000.00	1,850,000.00	2,990,000.00	-19.12%
221	627	Thorax	Ship	9,309,000.00	11,890,000.00	8,666,000.00	11,180,000.00	-4.28%
222	591	Tormentor	Ship	450,100.00	688,700.00	100,600.00	589,800.00	5.09%
223	593	Tristan	Ship	480,000.00	534,700.00	274,000.00	498,700.00	-3.50%
224	32880	Venture	Ship	286,700.00	348,100.00	205,500.00	299,600.00	4.57%
225	626	Vexor	Ship	10,010,000.00	11,980,000.00	9,553,000.00	12,070,000.00	-10.67%
226	33003	Enormous Freight Container	Structure	238,700.00	399,600.00	500.00	239,000.00	50.48%
227	24445	Giant Freight Container	Structure	133,800.00	274,000.00	50,590.00	512,000.00	-51.84%

Any cell with a negative number is a loss, a positive number is a profit opportunity if you want to take the risk. You need to examine the data carefully. Other players will do tricks in the market such as placing very low buy orders to drive prices down and trick unsuspecting newbies into selling immediate for very low prices. Sometimes they will post unrealistic sell prices to trick unsuspecting payers into paying too much for items, or to drive prices up.

Some cells with show #VALUE. This indicates that one of the prices needed for the profit calculation is missing. No order of the type we are looking for exists on the market if it says “No Price.”



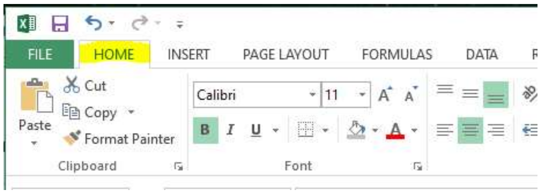
	ItemI	Item Name	Class	Jita Buy	Jita Sell	DDX Buy	DDX Sell	Jita->DDX
123	62528	Compressed Plagioclase	Ore	49.03	51.28	33.13	50.00	-12.25%
124	62530	Compressed Rich Plagioclase	Ore	46.21	54.00	32.40	46.90	-21.83%
125	62520	Compressed Scordite	Ore	16.00	16.50	10.11	17.90	-2.36%
126	62460	Compressed Sylvite	Ore	790.10	1,035.00	712.10	No Price	#VALUE!
127	62516	Compressed Veldspar	Ore	11.03	16.00	9.01	19.90	11.94%

If you want to make a profit using immediate sales, you need to change the formula based on what you have learned. Change the formula in cell “K2” to “=(($H2*0.9)/$F2)-1” and copy it down to all rows with item data.

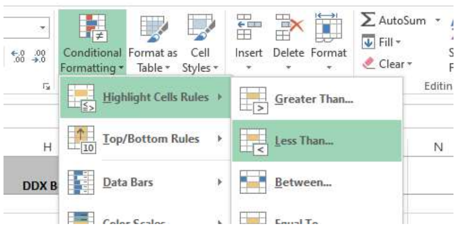
To make it easier to quickly identify what is profitable and what is not, you can use conditional formatting to color losses as red. Use the instructions following for the spreadsheet product you are using.

Conditional formatting – Excel

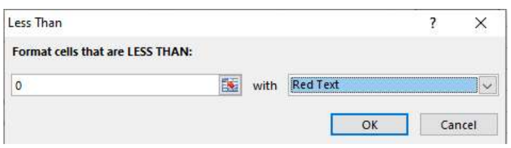
Make sure you have the “Home” tab selected on the ribbon bar.



Select the entire column “K” and then click the Conditional Formatting icon on the ribbon bar and choose “Highlight Cells Rules”, “Less Than…”



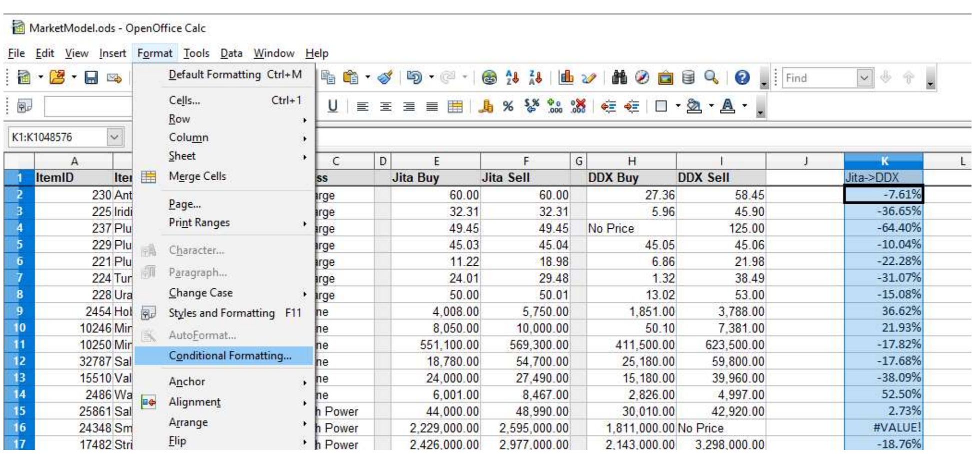
Use the options below:



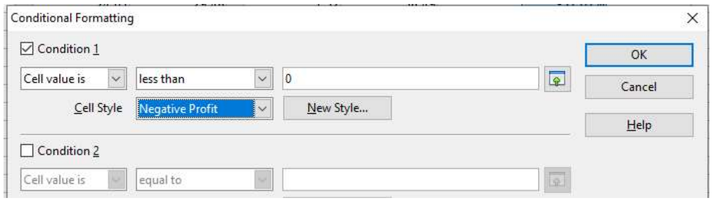
Any loss will now show as red.

Conditional formatting - OpenOffice

Select the entire column “K” and from the menu select “Format”, “Conditional Formatting…”



Select the options below:



Any loss will now show as red.

ItemID	Item Name	Class	Jita Buy	Jita Sell	DDX Buy	DDX Sell	Jita->DDX
230	Antimatter Charge M	Charge	60.00	60.00	27.36	58.45	-7.61%
225	Iridium Charge M	Charge	32.31	32.31	5.96	45.90	-36.65%
237	Plutonium Charge L	Charge	49.45	49.45	No Price	125.00	-64.40%
229	Plutonium Charge M	Charge	45.03	45.04	45.05	45.06	-10.04%
221	Plutonium Charge S	Charge	11.22	18.98	6.86	21.98	-22.28%
224	Tungsten Charge M	Charge	24.01	29.48	1.32	38.49	-31.07%
228	Uranium Charge M	Charge	50.00	50.01	13.02	53.00	-15.08%
2454	Hobgoblin I	Drone	4,008.00	5,750.00	1,851.00	3,788.00	36.62%
10246	Mining Drone I	Drone	8,050.00	10,000.00	50.10	7,381.00	21.93%

Ore sheet – "Miner" model (Adding ore and minerals)

The Ore sheet is designed to answer the question of "how do I make the most from what I mine?"

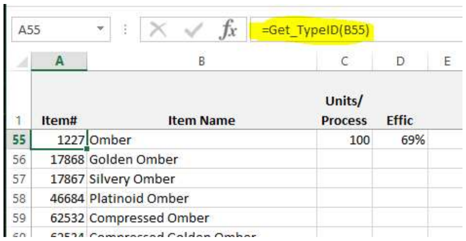
It has 5 parts:

- Information (Green) – Item ID, Name and data needed for reprocessing calculations.
- Market Buy/Sell (Blue) – What is the item selling and being bought for in the market?
- Compressed Markey Buy/Sell (Purple) – If it can be compressed, what are the compressed item market buy and sell prices?
- Refined Market Buy/Sell (Yellow) – If I refine it to minerals, what is their combined buy or sell price?
- Refined results (Red) – Data required for the refined market price, if I refine it, what does it make?

Item#	Item Name	Units/ Process	Effic	Jita Buy	Jita Sell	Jita Buy Compressed	Jita Sell Compressed	Jita Buy Minerals	Jita Sell Minerals	ritanium	Mexallon	Pyerite	Oxygen Isotopes	Helium Isotopes	Hydroge Isotope
18	Plagioclase	100	69%	38.53	43.00	40.50	43.00	38.63	40.98	175	70				
17455	Azure Plagioclase	100	69%	37.57	37.58	46.68	47.00	40.81	43.29	184	74				
17456	Rich Plagioclase	100	69%	40.23	45.00	48.26	50.99	42.51	45.09	193	77				
1228	Scordite	100	69%	14.38	17.00	17.26	22.00	9.10	9.64	150		90			
17463	Condensed Scordite	100	69%	14.38	15.25	15.71	17.70	9.60	10.17	158		95			
17464	Massive Scordite	100	69%	13.25	17.75	17.60	22.00	10.01	10.60	165		99			
1230	Veldspar	100	69%	9.96	12.50	11.87	13.55	10.46	10.98	400					
17470	Concentrated Veldspar	100	69%	10.54	10.63	12.32	13.75	10.98	11.53	420					
17471	Dense Veldspar	100	69%	11.15	11.16	13.89	14.74	11.51	12.08	440					
16264	Blue Ice	1	69%	1,401.00	152,000.00	136,300.00	147,400.00	132,833.66	147,022.23				414		
28433	Compressed Blue Ice	1	69%	136,300.00	147,400.00	---	---	132,833.66	147,022.23				414		

There will be cases where, say for ice or low sec ore, they need to be added to the model. For this example, we will use Omber. There's 4 types of Omber, and then there are the compressed versions to add. If you have some old compressed ore, you will also need to add the batch compressed versions of the ore as well.

You can add the formula “=Get_TypeID()” to verify that everything in the item name is spelled correctly. If it isn’t, the Type ID will return a -1 instead of the correct ID.



	A	B	C	D	E
1	Item#	Item Name	Units/ Process	Effic	
55	1227	Omber	100	69%	
56	17868	Golden Omber			
57	17867	Silvery Omber			
58	46684	Platinoid Omber			
59	62532	Compressed Omber			

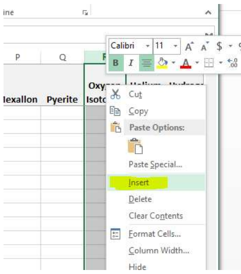
Using the instructions provided earlier, add the ores to the Items sheet and the WatchItems.csv.

You can select the ore in the regional market, right-click on it to “Show Info” and click the “Attributes” tab. You can then find the value for the “Units/Process” column.

The value for the reprocessing efficiency can be obtained when you are in your favorite refinery. Right-click on some ore and select “Reprocess” and the efficiency will be shown in the dialog.

In this case, Omber refines into Pyerite and Isogen. Isogen is not listed in the Ore sheet, so it needs to be added to the “refined results” section. It doesn’t need to be added elsewhere, it’s already in the Items sheet.

To make it really easy, just insert a column after the “Pyerite” column. This way the formulas will “adjust” to include the inserted column.



Enter how many Pyerite and Isogen result from the refining.

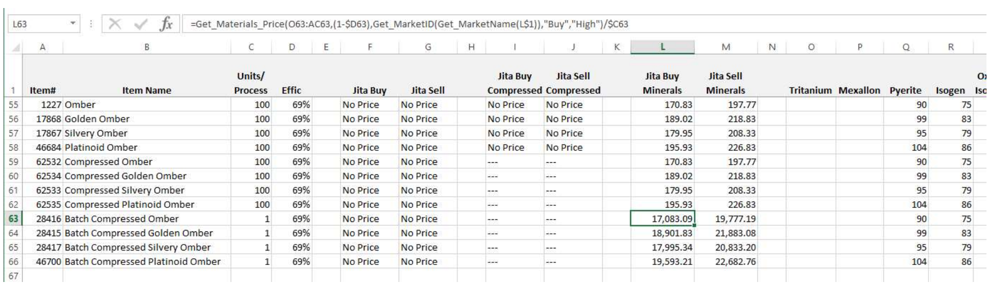
	Item#	Item Name	Units/ Process	Effic	Jita Buy	Jita Sell	Jita Buy Compressed	Jita Sell Compressed	Jita Buy Minerals	Jita Sell Minerals	Tritanium	Mexallon	Pyerite	Isogen
55	1227	Omber	100	69%									90	75
56	17868	Golden Omber												
57	17867	Silvery Omber												

Repeat this for each of the ores listed.

Make sure that the formula for the “Sell Minerals” has all the result columns selected, in this case outlined in blue. If not, adjust the formulas for the “Buy Minerals” and “Sell Minerals” columns and make sure they are all updated.

	Jita Sell Minerals	Tritanium	Mexallon	Pyerite	Isogen	Oxygen Isotopes	Helium Isotopes	Hydrogen Isotopes	Nitrogen Isotopes	Liquid Ozone	Heavy Water	Strontium Clathrates	Hydrocarbons	Silicates	Evaporite Deposits	Atmospheric Gases
12	2,210.57		800	4000										130		
53	2,290.88		800	8000											130	
84	ll","Low")/$C5		800	16000												130
				90	75											
				99	83											
				95	79											
				104	86											

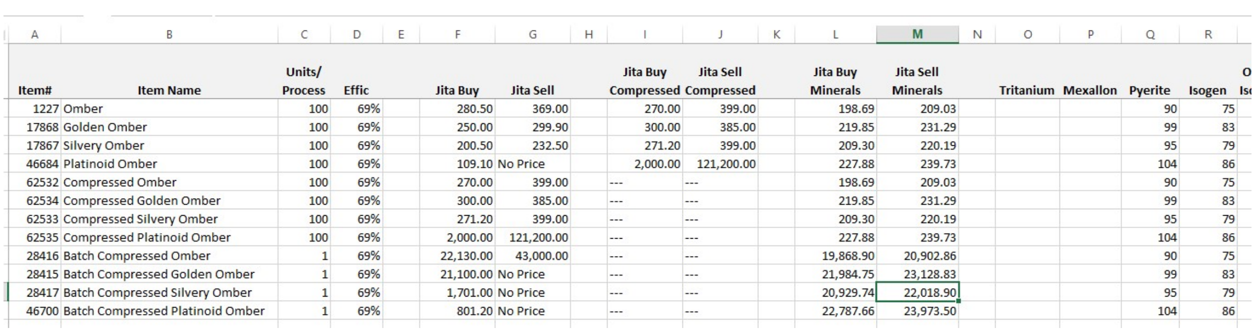
Copy and paste the formulas down in each column.



L63 =Get_Materials_Price(O63:AC63,(1-$D63),Get_MarketID(Get_MarketName(L$1)),"Buy","High")/$C63

	Item#	Item Name	Units/ Process	Effic	Jita Buy	Jita Sell	Jita Buy Compressed	Jita Sell Compressed	Jita Buy Minerals	Jita Sell Minerals	Tritanium	Mexallon	Pyerite	Isogen
55	1227	Omber	100	69%	No Price	No Price	No Price	No Price	170.83	197.77			90	75
56	17868	Golden Omber	100	69%	No Price	No Price	No Price	No Price	189.02	218.83			99	83
57	17867	Silvery Omber	100	69%	No Price	No Price	No Price	No Price	179.95	208.33			95	79
58	46684	Platinoid Omber	100	69%	No Price	No Price	No Price	No Price	195.93	226.83			104	86
59	62532	Compressed Omber	100	69%	No Price	No Price	---	---	170.83	197.77			90	75
60	62534	Compressed Golden Omber	100	69%	No Price	No Price	---	---	189.02	218.83			99	83
61	62533	Compressed Silvery Omber	100	69%	No Price	No Price	---	---	179.95	208.33			95	79
62	62535	Compressed Platinoid Omber	100	69%	No Price	No Price	---	---	195.93	226.83			104	86
63	28416	Batch Compressed Omber	1	69%	No Price	No Price	---	---	17,083.09	19,777.19			90	75
64	28415	Batch Compressed Golden Omber	1	69%	No Price	No Price	---	---	18,901.83	21,883.08			99	83
65	28417	Batch Compressed Silvery Omber	1	69%	No Price	No Price	---	---	17,995.34	20,833.20			95	79
66	46700	Batch Compressed Platinoid Omber	1	69%	No Price	No Price	---	---	19,593.21	22,682.76			104	86

Update the prices following the documentation.



Item#	Item Name	Units/ Process	Effic	Jita Buy	Jita Sell	Jita Buy Compressed	Jita Sell Compressed	Jita Buy Minerals	Jita Sell Minerals	Tritanium	Mexallon	Pyerite	Isogen
1227	Omber	100	69%	280.50	369.00	270.00	399.00	198.69	209.03			90	75
17868	Golden Omber	100	69%	250.00	299.90	300.00	385.00	219.85	231.29			99	83
17867	Silvery Omber	100	69%	200.50	232.50	271.20	399.00	209.30	220.19			95	79
46684	Platinoid Omber	100	69%	109.10	No Price	2,000.00	121,200.00	227.88	239.73			104	86
62532	Compressed Omber	100	69%	270.00	399.00	---	---	198.69	209.03			90	75
62534	Compressed Golden Omber	100	69%	300.00	385.00	---	---	219.85	231.29			99	83
62533	Compressed Silvery Omber	100	69%	271.20	399.00	---	---	209.30	220.19			95	79
62535	Compressed Platinoid Omber	100	69%	2,000.00	121,200.00	---	---	227.88	239.73			104	86
28416	Batch Compressed Omber	1	69%	22,130.00	43,000.00	---	---	19,868.90	20,902.86			90	75
28415	Batch Compressed Golden Omber	1	69%	21,100.00	No Price	---	---	21,984.75	23,128.83			99	83
28417	Batch Compressed Silvery Omber	1	69%	1,701.00	No Price	---	---	20,929.74	22,018.90			95	79
46700	Batch Compressed Platinoid Omber	1	69%	801.20	No Price	---	---	22,787.66	23,973.50			104	86

T1 sheet – “Maker” model (Adding blueprints and materials)

T2 sheet – “Maker” model (Adding blueprints and materials)

Custom Functions

Get_Market_Price

Get_Materials_Price

Get_TypeID

Get_MarketID

Get_MarketName

Get_RegionID

Get_MakePrice